St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Musculoskeletal System | IMCQ 2

IMCQ 2

School of Medicine,

St. George's University

1. A 22-year-old man is brought to the emergency department because of a 2-hour history of severe pain radiating across his back and down his left upper limb. Imaging studies show an unusual fracture through the superior border of the left scapula extending superiorly through the suprascapular notch. Which of the following movements of the arm will most likely be affected during physical examination?

A. Adduction and internal rotation

B. Abduction from 15-90 degrees and internal rotation

C. Abduction from 15-90 degrees and external rotation

D. Abduction for the first 15 degrees and external rotation

E. Abduction for the first 15 degrees and internal rotation

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SOM.MKII.BPM1.1.MSK.1.ANAT.1149	List the attachments, innervations and actions of the muscles that act on the shoulder joint.

2. A 12-year-old boy is brought to the physician by his father because of a 2-day history of fever, fatigue, and right shoulder pain. Physical examination shows a swollen mass in the axillary region. Any movement at the shoulder joint increases the pain but movement testing of the elbow and wrist joints shows wrist-extension weakness and loss of triceps reflex. Sensory examination shows loss of sensation to light touch over the right upper lateral, lower lateral and posterior arm. Imaging study identifies an abscess in the right axillary region. Which of the following structures is most likely being compressed by the abscess to resulting in the patient's symptoms?

A. Axillary nerve

B. Lateral cord of the brachial plexus

C. Medial cord of the brachial plexus

D. Posterior cord of the brachial plexus

E. Radial nerve

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SOM.MKII.BPM1.1.MSK.1.ANAT.1134	List the fiber types of the nerves that are present in the branches of the brachial plexus.

3. An 18-year-old man is brought to the emergency department by paramedics because of a stab wound to the axilla. Physical examination shows the knife has penetrated the axillary artery, between the 2nd and 3rd parts. The anastomosis between which of the following arteries is most likely providing blood supply to the distal parts of the upper limb?

A. Suprascapular and thoracodorsal

B. Dorsal scapular and thoracodorsal

C. Dorsal scapular and circumflex scapular

D. Circumflex scapular and lateral thoracic

E. Suprascapular and posterior circumflex humeral

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E. Suprascapular and posterior circumflex humeral

SOM.MKII.BPM1.1.MSK.1.ANAT.1130	List the arteries involved in anastomosis around the scapula, give the origin artery for each and discuss the clinical importance of this anastomoses.

4. A 20-year-old man comes to the physician because of a 4-day history of episodic tingling and numbness in his right hand, that started after a practice baseball session. The episodes seem to have increased in frequency over the last couple of days. On examination, the patient's right wrist is mildly swollen and tender. Sensory testing shows decreased sensation to light touch on the medial palmar surface of the right hand and the right little finger. An x-ray of this patient's hand will most likely show a fracture of which of the following indicated structures?

A. A
B. B
C. C
D. D
E. E
F. F
G. G

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A. A
B. B
C. C
D. D
E. E
F. F
G. G

SOM.MKII.BPM1.1.MSK.1.ANAT.1182	Identify the following carpal bone fractures: Scaphoid fracture and (Fracture of hook of hamate) and describe their possible clinical consequences..

5. **A 25-year-old man comes to the physician for a follow-up examination. One month ago, he was stabbed in the lateral chest wall. Physical examination shows his scapula moves away from the thoracic wall when he leans on his right hand as shown in the photograph. Which of the following additional functional deficits will most likely be present in this patient?**

A. Inability to shrug the shoulder

B. Decreased extension of the shoulder joint

C. Difficulty in abducting his arm above shoulder level

D. Difficulty in flexing the shoulder joint

E. Difficulty in adducting the scapula

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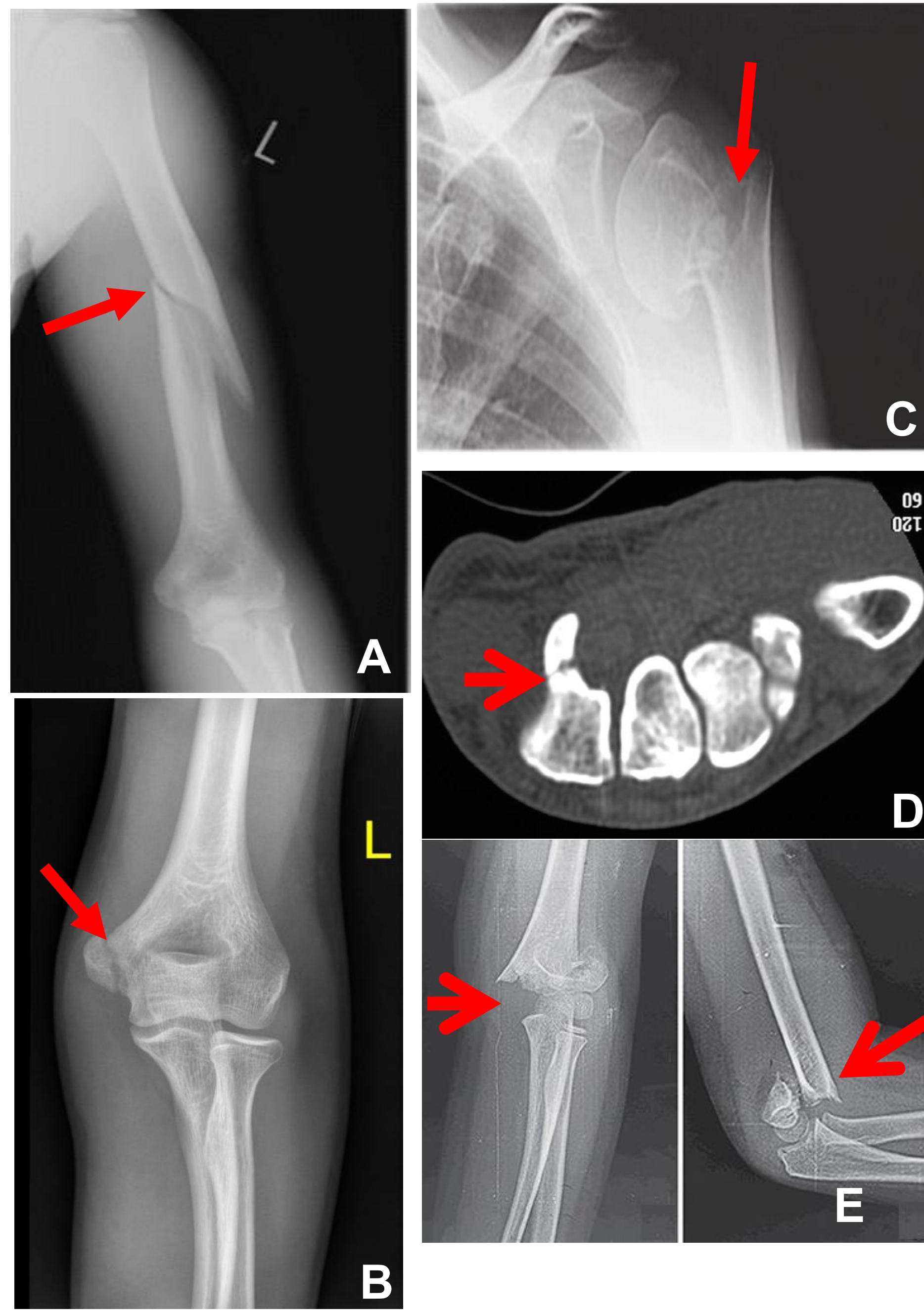
D. Difficulty in flexing the shoulder joint
E. Difficulty in adducting the scapula

SOM.MKII.BPM1.1.MSK.1.ANAT.1126	List the general attachments, innervations and actions of the muscles that contribute to the walls of the axilla.

6. A 6-year-old girl is brought to the emergency department following an injury to her right arm after she slipped and fell onto her outstretched right hand 1 hour ago. The patient had immediate pain and would not move her arm. She did not hit her head or lose consciousness. Physical examination shows decreased sensation over the mid-palmar aspect of the hand extending to the base of the index finger. Which of the following will most likely be found on imaging studies?

A. A
B. B
C. C
D. D
E. E



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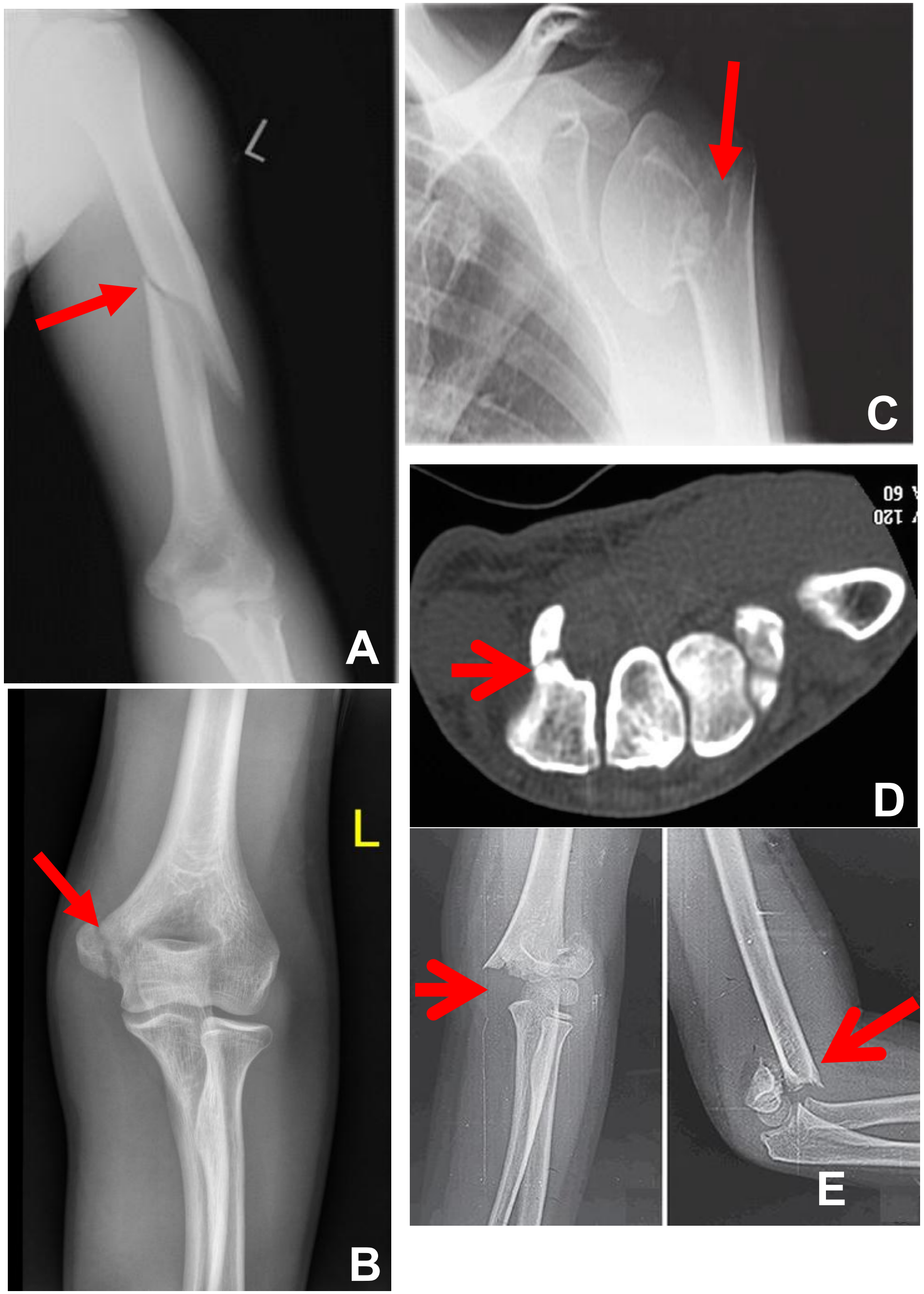

A. A
B. B
C. C
D. D
E. E

SOM.MKII.BPM1.1.MSK.1.ANAT.1162	Explain the potential clinical consequences of fractures at the surgical neck, midshaft, supracondylar, and medial epicondylar regions of the humerus and be able to recognize them on a radiograph.
SOM.MKII.BPM1.1.MSK.1.ANAT.1182	Identify the following carpal bone fractures: scaphoid fracture and fracture of hook of hamate and describe their possible clinical consequences.

7. A 52-year-old woman comes to the physician for a follow-up examination 3 months after she underwent surgery on the left breast. Physical examination shows difficulty adducting, medially rotating, and extending the left arm. This is most likely due to damage to a nerve during her surgery. Which of the following nerve roots most likely contribute to this injured structure?

A. C5 and 6
B. C6, 7 and 8
C. C8 and T1
D. C5 to T1
E. C5, C6 and C7

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C. C8 and T1
D. C5 to T1
E. C5, C6 and C7

SOM.MKII.BPM1.1.MSK.1.ANAT.1149	List the attachments, innervations and actions of the muscles that act on the shoulder joint.

8. A 52-year-old man comes to the physician because of a 2-month history of worsening shoulder pain. The pain is more pronounced when he attempts to lift objects overhead, such as retrieving files from high shelves in his office. There is no history of trauma or significant lifting activities. Physical examination shows tenderness on initiation of abduction of the right arm against resistance, with his elbows extended and thumbs pointed inferiorly. Which of the following nerves most likely innervates the affected muscle?

A. Dorsal scapular
B. Suprascapular
C. Upper subscapular
D. Lower subscapular
E. Long thoracic

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SOM.MKII.BPM1.1.MSK.1.ANAT.1126	List the general attachments, innervations and actions of the muscles that contribute to the walls of the axilla.

9. A 72-year-old man comes to the physician for a routine health maintenance examination. He has a history of uncontrolled hypertension for the past 10 years. He smoked one pack of cigarettes daily for 45 years. He reports that since last year, he has been unable to hold a cigarette like he used to. When asked to clarify, he points to an advertisement in an old magazine from the waiting room showing the following hand position. Physical examination shows he is unable to attain this position to hol a cigarette. Which of the following muscles would most likely be affected?

A. Flexor pollicis longus

B. Extensor digitorum

C. Palmar interossei

D. Adductor pollicis

E. Abductor digiti minimi

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SOM.MKII.BPM1.1.MSK.1.ANAT.1189	List the attachments, innervations and actions of the intrinsic muscles of the hand.

10. An 18-year-old man is brought to the emergency department by his friends because of a profusely bleeding wound to the axilla. Physical examination shows the knife has penetrated the axillary artery deep to the pectoralis minor muscle tendon. The patient has loss of sensation to the lateral forearm. Which of the following muscle actions will most likely be lost ?

A. Flexion of the arm
B. Extension of the arm
C. Flexion of the forearm
D. Extension of the forearm
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SOM.MKII.BPM1.1.MSK.1.ANAT.1151	List the general attachments, innervations, blood supply and actions of the muscles in the flexor and extensor compartments of the arm

11. A 30-year-old man comes to the emergency department following a high-speed motor vehicle collision. He complains of numbness and a tingling sensation along the lower medial side of his arm. Physical examination shows anesthesia over the lower medial aspect of his arm. No other functional deficits were noted. Imaging studies show injury to the spinal cord. Which of the following spinal cord levels is most likely injured resulting in the deficit shown in this patient?

A. C5
B. C6
C. C7
D. C8
E. T1
F. T2

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C. C7
D. C8
E. $\boxed{\text{T1}}$
F. T2

SOM.MKII.BPM1.1.MSK.1.ANAT.1132	Describe the organization of the brachial plexus from its roots to its terminal branches.
SOM.MKII.BPM1.1.MSK.1.ANAT.1134	List the fiber types of the nerves that are present in the branches of the brachial plexus.

12. A 43-year-old man comes to the physician for a follow-up examination. He had a work-related accident which involved the left upper extremity, 6 weeks ago. Physical examination shows inability to flex the distal phalanx of digits 2 and 3 but he can flex MCP and PIP joints of these fingers. He also has difficulty in pronating the forearm. Which of the following additional findings is most likely to be found in this patient?

A. Loss of sensation to the dorsum of the hand
B. Decreased extension of the digits
C. Decreased flexion of the DIP of digits 4 and 5
D. Decreased flexion of the IP joint of the thumb
E. Loss of sensation to the palmar 1 and ½ digits

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13. A 23-year-old man is brought to the emergency department following a motorcycle collision. He is unable to abduct his arm at the shoulder joint and has a loss of sensation over the lateral aspect of his arm. His right arm hangs by his side. Physical examination shows a medially rotated upper limb in extension with the palm facing posteriorly. An MRI of the brachial plexus shows a tear in nerve roots (avulsion injury). Which of the following roots of the brachial plexus were most likely injured resulting in these findings?

A. C8-T1
B. C3-C5
C. C5-C6-C7
D. C5-C6
E. C8-T2

13. A 23-year-old man is brought to the emergency department following a motorcycle collision. He is unable to abduct his arm at the shoulder joint and has a loss of sensation over the lateral aspect of his arm. His right arm hangs by his side. Physical examination shows a medially rotated upper limb in extension with the palm facing posteriorly. An MRI of the brachial plexus shows a tear in nerve roots (avulsion injury). Which of the following roots of the brachial plexus were most likely injured resulting in these findings?

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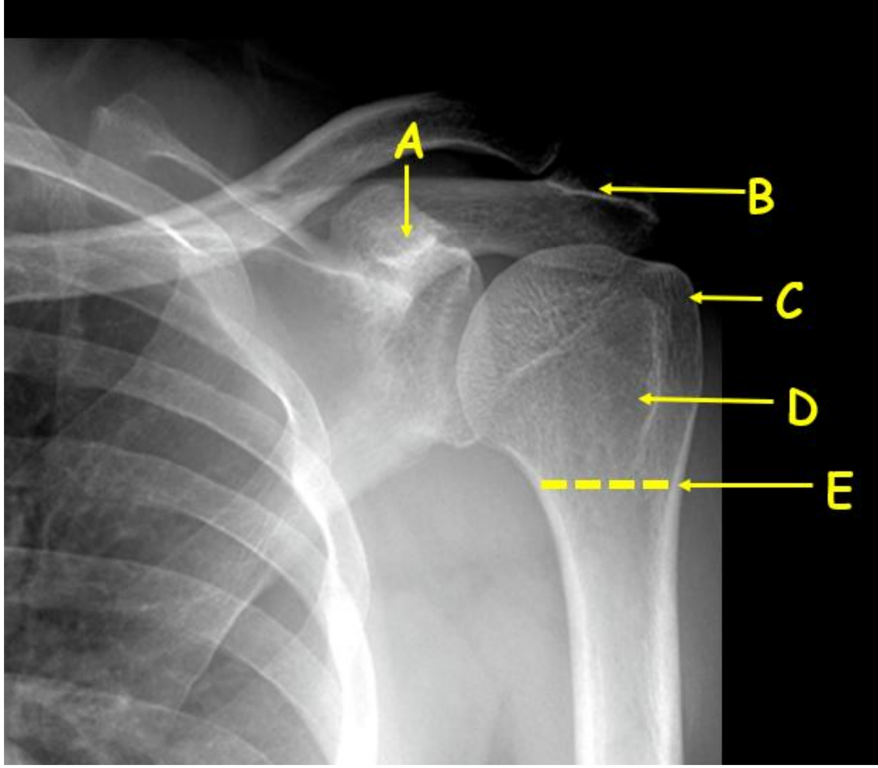
SOM.MKII.BPM1.1.MSK.1.ANAT.1134	List the fiber types of the nerves that are present in the branches of the brachial plexus.

14. A 17-year-old high school student is brought to the emergency department because of a 2-hour history of right shoulder pain. The pain started after he hit his shoulder on the ground while tackling another player during a rugby game. Physical examination shows redness and swelling in the right shoulder area. Movement testing is limited by the pain but sensory testing shows an area of diminished sensation in the upper lateral arm. A fracture of which of the following labeled areas could most likely damage the nerve supplying the area affected in this patient?



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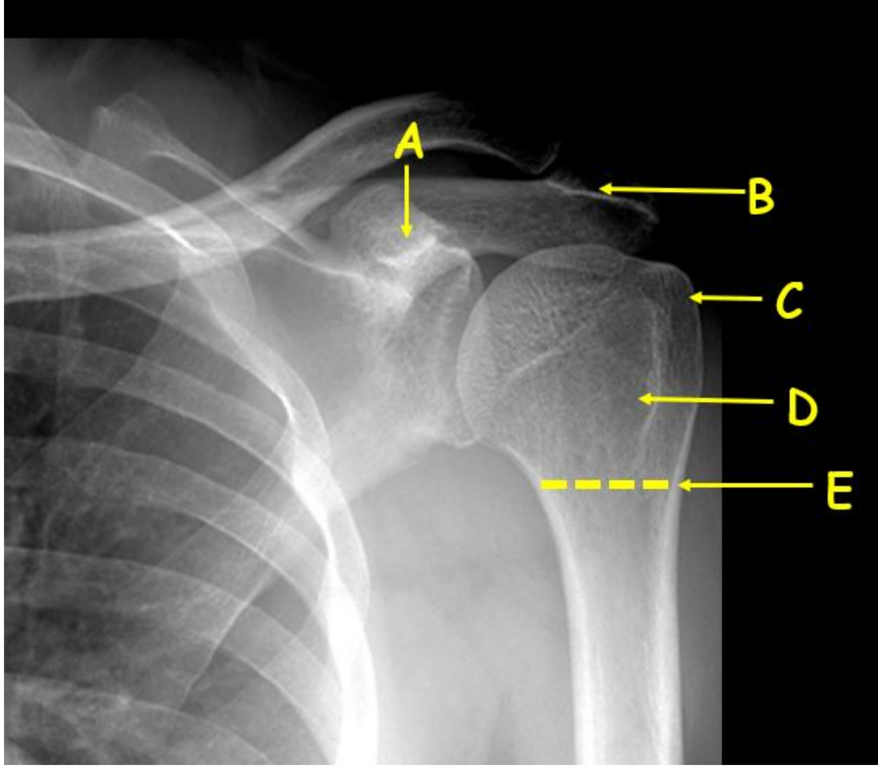

A. A

B. B

C. C

D. D

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SOM.MKII.BPM1.1.MSK.1.ANAT.1162 Explain the potential clinical consequences of fractures at the surgical neck, midshaft, supracondylar, and medial epicondylar regions of the humerus and be able to recognize them on a radiograph.

15. A 34-year-old man comes to the emergency department because of a cut to his left hand that he sustained 2 hours ago while using a saw as part of DIY home improvements. Physical examination shows a laceration over the palmar surface of the middle section of his left little finger. A subcutaneous local block is used to numb the area to allowing it to be sutured. The nerve that is being blocked in this case also provides sensory innervation to which of the following areas?

A. Palmar surface of lateral 3 ½ digits

B. Dorsal surface of lateral 3 ½ digits

C. Medial surface of the hand

D. Skin covering the anatomical snuff box

E. Skin overlying the adductor pollicis muscle

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E. Skin overlying the adductor pollicis muscle

SOM.MKII.BPM1.1.MSK.1.ANAT.1208 Describe the signs and symptoms of lesions of the ulnar nerve in the arm and wrist and palm.

16. An 18-year-old man is brought to the emergency department following a fall from a height. His pulse is 108/min, respirations are 26/min, blood pressure is 93/48 mm Hg, and temperature is 37°C (98.6°F). He complains of severe pain and inability to move in his left arm. Physical examination shows swelling, tenderness, and visible deformity in the left arm. An x-ray of his left upper limb is shown. Which of the following muscle actions will most likely be completely lost?

A. Flexion of the arm
B. Extension of the arm
C. Supination of the forearm
D. Pronation of the forearm
E. Flexion of the wrist
F. Extension of the wrist

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17. A 30-year-old woman comes to the physician because of a 1-month history of difficulty moving her right upper limb. Physical examination shows wrist drop and weakness of grasp but normal extension of the elbow joint. There is no loss of sensation in the affected limb. Which of the following nerves is most likely affected?

A. Radial

B. Anterior interosseous

C. Posterior interosseous

D. Median

E. Superficial radial

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D. Median

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1203 Describe the motor and sensory deficits of lesions of the radial nerve in the axilla, arm and wrist.